Mohammed Rawih

Senior Software Engineer – Full Stack Systems

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PROFESSIONAL SUMMARY

Senior Full Stack Software Engineer with over 7 years of hands-on experience delivering large-scale, mission-critical enterprise and government-grade systems within complex and high-impact environments in Saudi Arabia. Demonstrated expertise in owning the full software lifecycle — from deep requirements analysis and system architecture design to development from scratch, production deployment, monitoring, and long-term operation. Proven ability to build and operate highly reliable, scalable, and secure platforms that support seasonal peak loads, sensitive data, and operational continuity. Experienced in developing comprehensive ERP and operations systems, integrating applied artificial intelligence capabilities, and delivering solutions that align with real business, regulatory, and national-scale requirements. Known for strong technical depth, accountability, and consistent delivery of stable, maintainable systems trusted in production.

SKILLS

Core Technologies (Production Experience)

- **.NET 8/9** (C#, ASP.NET Core, EF Core, Minimal APIs)
- **Java** (Spring Boot, RESTful services, enterprise applications)
- **Python** (FastAPI, Django)
- **React.js & Next.js 15** (App Router, Server Components)

Data Platforms & Storage

- **PostgreSQL** (enterprise-grade systems, heavy indexing, partitioning, and performance tuning)
- **SQL Server** (large-scale ERP systems, stored procedures, replication)
- **MySQL / MariaDB** (high-traffic systems with sharding experience)
- **Redis** (distributed caching, pub/sub, session and state management)
- **Supabase** (Authentication, Realtime, and Storage in production environments)

System Architecture & Performance

- Clean Architecture and Vertical Slice design
- Domain-Driven Design (DDD), CQRS, MediatR, Event-Driven architectures
- Advanced data modeling and query optimization, reducing query latency from seconds to milliseconds in multiple production systems
- Microservices architecture and high-performance API design for large-scale systems

DevOps & Cloud Infrastructure

- Docker and Kubernetes for containerized, scalable deployments
- CI/CD pipelines using GitHub Actions and GitLab CI
- Hands-on experience with **Alibaba Cloud** and **Microsoft Azure** in real production environments

Security, Reliability & Quality

- Secure API design using JWT, OAuth2, RBAC, and rate limiting
- Implementation of access control and permission management for enterprise systems
- Test-driven development (TDD), unit and integration testing using xUnit

Applied Artificial Intelligence

- Training and fine-tuning machine learning models for applied, real-world use cases
- Experience with **Large Language Models (LLMs)**, including prompt design, fine-tuning concepts, and production integration
- Design and implementation of **Retrieval-Augmented Generation (RAG)** pipelines using vector databases
- Image processing and computer vision workflows using CNN-based models
- Practical use of classical machine learning algorithms (e.g., KNN) for classification and pattern recognition
- Integration of AI capabilities into backend services and enterprise platforms via APIs

WORK EXPERIENCE

Full Stack Developer

Maham IT Solutions, Riyadh, Saudi Arabia **Feb 2024 – Present**

- Developed key modules of the Maham-AI ERP system (HR, CRM, Services) and integrated AI-driven analytics to enhance automation and decision-making.
- Designed and implemented PostgreSQL relational databases, ensuring scalability, data integrity, and optimized query performance.
- Implemented full-stack features using React and .NET Web API, fully integrated with Supabase services for database, authentication, and real-time synchronization.
- Improved system performance through SQL optimization, Redis caching, and backend refactoring, reducing response time and server load.
- Implemented a complete Role-Based Access Control (RBAC) system, enabling secure permissions and multi-level user access.
- Managed GitHub repositories, enforced code standards, and maintained CI/CD pipelines for continuous delivery.
- Deployed and monitored production environments on Alibaba Cloud, ensuring high availability, security, and reliable performance.

Backend Developer

HowbaniSoft Riyadh, Saudi Arabia(remotely) **Dec 2020 – Jun 2023**

- Designed, developed, and maintained backend systems using .NET ,Java technologies.
- Built, optimized, and managed SQL Server databases to improve performance and reliability.
- Developed secure and scalable RESTful APIs for various business applications.
- Implemented Docker to containerize services and streamline deployment workflows.
- Utilized GitHub for version control, code reviews, and collaborative development.
- Contributed to system architecture improvements and backend performance optimization.
- Worked efficiently under pressure, delivering high-quality backend solutions that enhanced the company’s technical operations.

Full-Stack web and Mobile Developer

Al-Rahal for IT Solutions And Web Services, Riyadh, Saudi Arabia (Freelance) **Jan 2018 – May 2019**

- Built and deployed user-friendly and responsive web applications using HTML, CSS, JavaScript, and React.js.

- Designed and implemented scalable and secure back-end systems utilizing .NET, and Django frameworks.
- Created robust Fast APIs to facilitate efficient data exchange between front-end and back-end applications.
- Led code reviews to ensure code quality and adherence to coding standards and best practices.
- Collaborated closely with cross-functional teams (e.g., marketing and designers) to analyze requirements, design solutions, and ensure successful project implementation.

Key Projects

OneMay – Enterprise ERP & Operations Platform (Boulevard & Riyadh Seasons)

- Led the development of a large-scale, **enterprise-grade ERP and operations platform** designed to support **Boulevard management and Riyadh Seasons–scale events**.
- Delivered a unified system combining **core ERP capabilities** with advanced **event and crowd operations**, built to operate in high-traffic, mission-critical environments.
- Implemented comprehensive ERP modules covering **Finance, Human Resources, Customers, Vendors, Contracts, Procurement, and Operational Reporting**.
- Designed and developed specialized systems for **event management, gate and access control, crowd management, risk management, and operational coordination**.
- Built the platform using **Next.js, React, and TypeScript** on the frontend, with **.NET 8 / ASP.NET Core** powering backend services.
- Developed and integrated **30+ enterprise modules** supporting both administrative and on-ground operational workflows.
- Designed and implemented **200+ RESTful APIs** and **100+ production-ready screens**, supporting complex, role-driven business processes.
- Established a scalable frontend architecture using **reusable components and a monorepo (Turborepo)**, reducing development time by approximately **70%**.
- Successfully migrated the frontend from **Vue 3 to React / Next.js** while preserving full system functionality and operational stability.
- Implemented critical features including **full Arabic/English bilingual support with RTL**, **QR-based ticketing**, **role-based access control (RBAC)**, and secure payment gateway integrations.
- Optimized data models and performance using **PostgreSQL and SqlSugar**, supported by **Redis caching** to ensure low-latency operation under peak seasonal traffic.
- Designed the system to meet **high availability, performance, and reliability requirements** expected in large-scale government and public-sector environments.

Legatoo AI platform for analyze the legal cases

- Developed an AI-driven legal consulting platform tailored specifically for **Saudi legal regulations and practices**.
- Built a web-based system for **contract analysis and legal case review**, capable of identifying ambiguous, conflicting, or non-compliant clauses.
- Implemented **Retrieval-Augmented Generation (RAG)** pipelines to enable context-aware legal question answering based on Saudi laws and regulations.
- Integrated AI capabilities to support **legal consultation workflows**, providing actionable insights aligned with local legal frameworks.
- Enhanced accuracy and efficiency in legal document validation, significantly reducing manual review effort for legal professionals.
- Designed the platform to function as both a **legal analysis tool** and an **interactive legal inquiry system** for lawyers and consultants.

Maham-AI ERP – AI-Powered Enterprise Resource Planning System

- AI-powered ERP platform designed to automate enterprise operations across HR, CRM, and service/location management. Built using .NET , React, and Supabase with PostgreSQL.
- Developed key operational modules including HR Management System and Boulevard Rental Tracking System for managing shops, leases, and service areas
- Designed and implemented a scalable database architecture (PostgreSQL) ensuring data integrity, modular structure, and optimized relational models
- Built responsive and interactive frontend interfaces using React, fully integrated with Node.js APIs
- Implemented a complete Role-Based Access Control (RBAC) system enabling granular permissions and secure access across all modules
- Enhanced system performance through SQL query optimization, caching strategies, and backend refactoring
- Collaborated within an 8-member team to design, develop, and deliver core ERP functionalities and AI-driven features
- Integrated AI capabilities using OpenAI API for automated insights, data analysis, and workflow assistance
- Leveraged Supabase for authentication, storage, and real-time features, enabling a modern cloud-driven architecture

Ra'ayati Secure, Scalable EHR Platform

- Contributed to the development of “Ra'ayati”, a unified national Electronic Health Record (EHR) platform designed to centralize medical records for citizens.
- Participated in designing a highly scalable and secure architecture supporting nationwide healthcare entities.
- Implemented robust user and permission management with strict access control to protect sensitive medical data.
- Supported complex multi-role healthcare workflows, including doctors, nurses, pharmacy staff, and administrators.
- Enabled interoperability between public and private healthcare providers to improve treatment accuracy and continuity of care.
- Supported full electronic healthcare operations, including diagnoses, prescriptions, and longitudinal medical history management.

EDUCATION

Bachelor of **Engineering in Telecommunications and Information Technology**
University of Taiz , GPA: 83.35% (Ranked 5th in Class) **Graduated: 2017**

CERTIFICATIONS

- Web Application Design and Development, Global Human Development and Research Center.
- Field Training, SabaFon, Telecom, Nov 2023 – Dec 2023
- CCNA Routing and Switching Essentials, 1 Apr 2019

LANGUAGES

- Arabic (Native)
- English (Proficient)

ADDITIONAL INFO

- Iqama (Valid for 2 Years)